LayoverNow — Semester Risk Management Report

Chrome Extension Project | 8-Week Semester | Solo Developer

1. Executive Summary

This report documents the comprehensive risk management process for LayoverNow, a Chrome Extension designed to help travelers discover and optimize stopover flights. The project was developed over an 8-week semester by a solo developer. The risk management process followed the Therac-25 risk analysis methodology, identifying risks by category, assessing probability and impact, and tracking mitigation progress week-by-week.

Six major risks were identified and analyzed across five categories: External Dependencies, Technical Architecture, Project Management, Security & Compliance, and User Experience. Each risk was decomposed into Q1 (Desired Behavior), Q2 (Preventative Behavior), and Q3 (Responsive Behavior) requirements, with detailed functional requirements derived for each quadrant.

2. Project Metadata

Project Name	LayoverNow
Type	Chrome Browser Extension (Manifest V3)
Duration	8 Weeks (Full Semester)
Team Size	1 Developer (Solo Project)
Repository	https://github.com/pranavapp2024-sys/LayoverNow

3. Risk Assessment Methodology

Risks are evaluated using a standard Probability x Impact matrix. Each dimension uses a 5-point Likert scale:

- Probability (P): 1=Rare, 2=Unlikely, 3=Possible, 4=Likely, 5=Almost Certain
- Impact (I): 1=Insignificant, 2=Minor, 3=Moderate, 4=Major, 5=Catastrophic

Risk Score = P x I. Classification:

- Low (1-5): Acceptable, monitor periodically
- Medium (6-12): Manage actively, implement mitigation

- High (15-20): Immediate action required, senior review
- Critical (21-25): Project-threatening, escalate immediately

4. Risk Categories

Technical / Architecture (T): Code quality, system design, performance, technology stack limitations

External Dependency (E): Third-party APIs, services, data sources, platform policies

Project Management (P): Scope, scheduling, resource availability, feature prioritization

Security & Compliance (S): User data privacy, regulatory compliance, secure data handling

User Experience (U): Usability, accessibility, device performance, user adoption

5. Risk Register Summary

The following table summarizes all identified risks at project initiation (Week 1):

Risk ID	Capability	Undesirable Event	Category	Risk Score	Priority
R1	Discover Stopover Flights	Flight API Unavailable/ Rate-Limited	E	20	Critical
R2	Interactive Flight Radar Map	Canvas Performance Degradation	U	12	High
R3	AeroAI Stopover Assistant	LLM API Failure/Excessive Latency	E	16	High
R4	Extension Installation	CWS Policy Violation/Rejection	E	12	High
R5	User Data Privacy	GDPR/CCPA Privacy Violation	S	12	High
R6	Project Delivery	Feature Overload/Scope Creep	P	15	High

6. Detailed Risk Analysis

6.1. Risk R1: Flight API Unavailable or Rate-Limited

Capability: Discover Stopover Flights

Risk Description: The third-party flight search API becomes unavailable, returns rate-limit errors, or changes its endpoint/schema. If the system cannot retrieve real-time flight data, users will see empty results or stale prices, rendering the extension useless.

Likelihood	Impact	Risk Score	Priority
4	5	20	Critical

Justification for Likelihood: Free-tier flight APIs are notoriously unstable. Skyscanner has strict rate limits (100 calls/day), Amadeus test environment has downtime, and no SLA guarantees exist for student projects.

Justification for Impact: The entire core value proposition depends on real-time flight data. Without API access, the extension cannot display stopover deals or calculate savings. Users would uninstall immediately.

Mitigation Strategy: 1. Multi-tier API fallback: Primary (Amadeus) -> Secondary (Skyscanner) -> Tertiary (static mock dataset). 2. Local caching layer with 4-hour TTL using chrome.storage.local. 3. Abstract API calls behind a FlightDataService interface. 4. Deploy a Node.js/Express proxy on Render to handle CORS and API key security.

Q1 - Desired Behavior (R1)

The system shall accept origin/destination airport codes, query the flight search API via proxy, retrieve real-time stopover options with pricing, filter by Best Value/Savings/Minimum Detour, and display top deals with route visualization.

Q2 - Preventative Behavior (R1)

The system shall not render deals without verifying valid API response fields. Shall not cache deals beyond 4 hours without refresh. Shall not make direct API calls from content script. Shall not store API keys in client-side code.

Q3 - Responsive Behavior (R1)

When API returns 429, 503, or timeout, serve cached results with stale-data banner, queue background retry, and display mock deals if no cache exists. Log all failures for debugging.

Functional Requirements - Q1 (R1)

- FR-1.1.1: FlightSearchModule shall accept user-input origin and destination IATA codes
- FR-1.1.2: FlightSearchModule shall construct and send search request to ProxyServer
- FR-1.1.3: FlightSearchModule shall parse JSON response and extract flight segments
- FR-1.1.4: DealFilterModule shall categorize results into Best Value, Savings, and Minimum Detour
- FR-1.1.5: UIRendererModule shall display filtered deals in scrollable card list
- FR-1.1.6: RadarMapModule shall plot flight path on canvas radar map.

Preventative Requirements - Q2 (R1)

- FR-1.2.1: DataValidationModule shall verify API responses contain non-null price, duration, and airport code fields before rendering
- FR-1.2.2: CacheManagerModule shall enforce maximum cache TTL of 4 hours
- FR-1.2.3: SecurityModule shall ensure all external API requests route through ProxyServer; no API keys in client scripts
- FR-1.2.4: RateLimitModule shall track API call counts per hour and throttle before exceeding quota
- FR-1.2.5: ProxyServer shall validate all incoming requests and reject unauthorized parameters.

Responsive Requirements - Q3 (R1)

- FR-1.3.1: ErrorHandlerModule shall detect HTTP 429, 503, 504, and network timeouts
- FR-1.3.2: CacheManagerModule shall retrieve most recent cached results when API error occurs
- FR-1.3.3: UIRendererModule shall display yellow 'Data may be stale' banner above cached results

- FR-1.3.4: FallbackModule shall load static mock data when both API and cache are unavailable
- FR-1.3.5: BackgroundRetryModule shall schedule automatic API retry after 5 minutes
- FR-1.3.6: AuditLogModule shall record all API failures with error code, timestamp, and route parameters.

6.2. Risk R2: Canvas Performance Degradation on Low-End Devices

Capability: Interactive Flight Radar Map

Risk Description: The custom canvas-based radar map performs poorly on low-end devices, causing frame rates below 15fps, UI freezing, or excessive battery drain. Users cannot visualize flight routes, leading to frustration.

Likelihood	Impact	Risk Score	Priority
4	3	12	High

Justification for Likelihood: The radar map uses continuous canvas rendering with bezier curves and animated dots. Many target users access Chrome on institutional Chromebooks with 4GB RAM and weak GPUs.

Justification for Impact: While poor performance degrades UX, it does not crash the extension. Users can still access the deal list. However, the radar map is a key differentiator, and poor performance undermines the premium brand.

Mitigation Strategy: 1. Delta-time scaling in animation loop. 2. Batch draw calls: render static background to offscreen canvas, redraw only moving elements. 3. 'Low Power Mode' toggle disabling animations and reducing to 10fps. 4. Auto-enable Low Power Mode on devices with <4GB RAM via navigator.deviceMemory. 5. Debounce resize events and use requestAnimationFrame exclusively.

Q1 - Desired Behavior (R2)

The system shall render an interactive radar map on HTML5 canvas, plotting origin, destination, and stopover waypoints with animated flight dots along bezier curve paths. The map shall support zoom and pan gestures.

Q2 - Preventative Behavior (R2)

The system shall not redraw the entire canvas on every frame. Shall not use setInterval for animation. Shall not enable full animation on devices with insufficient memory. Shall not block the main UI thread during rendering.

Q3 - Responsive Behavior (R2)

When frame rendering exceeds 33ms for 3 consecutive frames, automatically degrade to Low Power Mode: disable bezier curves, use straight lines, reduce to 10fps, and show 'Performance mode active' indicator. Allow manual override in settings.

Functional Requirements - Q1 (R2)

- FR-2.1.1: RadarMapModule shall initialize HTML5 canvas with 2D context
- FR-2.1.2: MapRendererModule shall draw world map background and airport markers using pre-rendered offscreen layers
- FR-2.1.3: PathDrawingModule shall calculate bezier control points for routes

- FR-2.1.4: AnimationModule shall animate flight dot using requestAnimationFrame with delta-time scaling
- FR-2.1.5: InteractionModule shall support mouse drag for panning and scroll wheel for zooming
- FR-2.1.6: LayerManagerModule shall composite static and dynamic layers each frame.

Preventative Requirements - Q2 (R2)

- FR-2.2.1: RenderOptimizerModule shall cache static map background to offscreen canvas, redraw only on resize
- FR-2.2.2: AnimationSchedulerModule shall use exclusively requestAnimationFrame; setInterval prohibited
- FR-2.2.3: PerformanceGateModule shall query navigator.deviceMemory and auto-enable Low Power Mode if <4GB
- FR-2.2.4: ThreadSafetyModule shall ensure canvas operations do not block UI event handlers
- FR-2.2.5: FrameBudgetModule shall monitor render time and trigger degradation if 33ms budget exceeded for 3 frames.

Responsive Requirements - Q3 (R2)

- FR-2.3.1: PerformanceMonitorModule shall measure time between requestAnimationFrame callbacks
- FR-2.3.2: DegradationEngineModule shall switch from bezier to straight lines in Low Power Mode
- FR-2.3.3: DegradationEngineModule shall reduce target frame rate from 60fps to 10fps
- FR-2.3.4: UINotificationModule shall display 'Performance mode active' badge when auto-degradation occurs
- FR-2.3.5: SettingsOverrideModule shall allow manual Low Power Mode toggle via chrome.storage.sync
- FR-2.3.6: RecoveryModule shall re-test performance every 10 seconds and restore full animation if frame times improve.

6.3. Risk R3: LLM API Failure or Excessive Latency

Capability: AeroAI Stopover Assistant

Risk Description: The AeroAI chatbot relies on third-party LLM API for generating personalized layover itineraries and visa guidance. If the API experiences downtime, quota exhaustion, or latency exceeding 10 seconds, the chatbot becomes non-functional.

Likelihood	Impact	Risk Score	Priority
4	4	16	High

Justification for Likelihood: LLM APIs frequently experience rate limiting on free tiers, unexpected downtime, and model deprecation. Student API keys have strict token limits that exhaust quickly with rich travel prompts.

Justification for Impact: The AeroAI assistant is a major feature differentiator. Users expecting instant AI guidance will be disappointed by a broken chatbot. The extension remains usable for deal discovery, but satisfaction drops significantly.

Mitigation Strategy: 1. 10-second request timeout with exponential backoff. 2. Comprehensive static tip database covering 50 common layover cities. 3. Cache successful AI responses keyed by route. 4. Graceful degradation: show static tip cards with 'AI is busy' message. 5. Use lightweight model (GPT-3.5-turbo) for faster, cheaper responses.

Q1 - Desired Behavior (R3)

The system shall provide an AI-powered chatbot accepting natural language queries about layover activities, visa, transport, weather, currency, and hotels. Shall generate contextual city-specific responses and display in conversational thread.

Q2 - Preventative Behavior (R3)

The system shall not send LLM requests exceeding 2000 tokens. Shall not block UI thread while waiting for AI response. Shall not display blank loading state for more than 2 seconds. Shall not cache AI responses longer than 24 hours.

Q3 - Responsive Behavior (R3)

When LLM API returns error, times out, or exceeds quota, immediately display pre-curated static tip cards relevant to the route with message: 'Our AI is taking a break - here are hand-picked tips for [City].' Log failure and queue background retry.

Functional Requirements - Q1 (R3)

- FR-3.1.1: ChatInterfaceModule shall render chat UI with input field and message history
- FR-3.1.2: QueryParserModule shall extract intent and entities from user messages
- FR-3.1.3: LLMServiceModule shall construct system prompt and send to LLM API via ProxyServer
- FR-3.1.4: ResponseFormatterModule shall parse LLM JSON and render formatted text

- FR-3.1.5: ContextManagerModule shall maintain conversation history for follow-up questions
- FR-3.1.6: MultiTopicModule shall handle visa, weather, currency, and transport queries.

Preventative Requirements - Q2 (R3)

- FR-3.2.1: TokenLimiterModule shall truncate or reject prompts exceeding 2000 tokens
- FR-3.2.2: AsyncRequestModule shall execute LLM calls asynchronously using fetch with AbortController
- FR-3.2.3: UXResponsivenessModule shall display static fallback within 2 seconds if AI not received
- FR-3.2.4: CacheExpiryModule shall enforce 24-hour TTL on cached AI responses
- FR-3.2.5: QuotaGuardModule shall track daily API calls and warn at 80% of free-tier quota.

Responsive Requirements - Q3 (R3)

- FR-3.3.1: ErrorDetectorModule shall catch HTTP 429, 500, 503, network errors, and timeout signals
- FR-3.3.2: FallbackContentModule shall retrieve static tips from static-tips.json based on stopover city and intent
- FR-3.3.3: UIMessageModule shall render fallback tips with 'AI is taking a break' header
- FR-3.3.4: BackgroundRetryModule shall schedule deferred retry using chrome.alarms
- FR-3.3.5: FailureLogModule shall record LLM failures with error type, timestamp, route, and prompt length
- FR-3.3.6: UserFeedbackModule shall allow rating AI responses and reporting incorrect information.

6.4. Risk R4: Chrome Web Store Policy Violation or Rejection

Capability: Extension Installation & Distribution

Risk Description: The Chrome Web Store may reject the extension due to excessive permissions, missing privacy policy, non-compliant code practices, or MV3 policy changes. Rejection prevents public distribution.

Likelihood	Impact	Risk Score	Priority
3	4	12	High

Justification for Likelihood: Chrome Web Store reviews have become stricter under MV3. Common rejections include broad permissions without justification, remotely hosted code, and missing privacy policies. However, careful preparation enables first-time approval.

Justification for Impact: Without Web Store distribution, the extension cannot reach real users or demonstrate market viability. For a semester project, sideloading is sufficient, but portfolio value is diminished. Rejection delays could miss the final demo.

Mitigation Strategy: 1. Request only minimum permissions: activeTab and storage. 2. Ensure all code bundled locally; no remote scripts or eval(). 3. Publish privacy policy on GitHub Pages. 4. Submit for review by Week 4 for 4-week buffer. 5. Prepare detailed screenshots and clear store descriptions.

Q1 - Desired Behavior (R4)

The system shall be packaged as a Chrome Extension compliant with Manifest V3, submitted to Chrome Web Store, and pass review for public distribution. Shall install cleanly without developer mode.

Q2 - Preventative Behavior (R4)

The system shall not request permissions beyond minimum required (activeTab, storage). Shall not execute remotely hosted code or use eval(). Shall not collect PII without explicit consent and published privacy policy.

Q3 - Responsive Behavior (R4)

If Web Store rejects extension, immediately deploy via developer mode sideloading for semester demo. Analyze rejection feedback, correct violations, and resubmit within 48 hours. Maintain compliance checklist.

Functional Requirements - Q1 (R4)

- FR-4.1.1: ManifestBuilderModule shall generate manifest.json with MV3 specs
- FR-4.1.2: BuildScriptModule shall bundle all JS/CSS/HTML with no external runtime dependencies
- FR-4.1.3: IconGeneratorModule shall create required icon sizes from master SVG
- FR-4.1.4: StoreSubmissionModule shall prepare screenshots and store description
- FR-4.1.5: PrivacyPolicyModule shall generate and host privacy policy HTML page.

Preventative Requirements - Q2 (R4)

- FR-4.2.1: PermissionAuditorModule shall review manifest permissions against whitelist [activeTab, storage, declarativeNetRequest]
- FR-4.2.2: CSPValidatorModule shall ensure CSP prohibits unsafe-eval and unsafe-inline
- FR-4.2.3: RemoteCodeScannerModule shall scan JS for eval(), new Function(), setTimeout(string) and fail build if found
- FR-4.2.4: DataCollectionGateModule shall require explicit user consent before storing data
- FR-4.2.5: PrivacyPolicyGateModule shall block submission until valid privacy policy URL is reachable.

Responsive Requirements - Q3 (R4)

- FR-4.3.1: SideloadPackageModule shall generate unpacked extension zip for chrome://extensions/ Developer Mode
- FR-4.3.2: RejectionAnalyzerModule shall parse CWS rejection emails for specific violation codes
- FR-4.3.3: ComplianceFixModule shall map violation codes to predefined fix procedures
- FR-4.3.4: ResubmissionModule shall re-package corrected extension and resubmit within 48 hours
- FR-4.3.5: ComplianceChecklistModule shall maintain living document of all CWS policies checked
- FR-4.3.6: FallbackDistributionModule shall provide sideloading instructions in README.md.

6.5. Risk R5: GDPR/CCPA Privacy Violation from Stored Search History

Capability: User Data Privacy & Compliance

Risk Description: The extension stores user search history and preferences in chrome.storage.local. If not properly anonymized, encrypted, or consented, the extension could violate GDPR or CCPA regulations.

Likelihood	Impact	Risk Score	Priority
3	4	12	High

Justification for Likelihood: Storing any user data triggers privacy obligations. Chrome Web Store requires privacy policies for extensions handling user data. A single complaint to Google could trigger review.

Justification for Impact: A privacy violation could lead to Web Store removal, academic integrity concerns, and personal liability. Privacy-by-design is a critical learning outcome and professional standard.

Mitigation Strategy: 1. Store ALL data in chrome.storage.local (on-device only). 2. Explicit consent flow on first install with privacy explanation and 'I Agree' checkbox. 3. 'Clear All Data' button in options. 4. Zero server-side logging on Render proxy. 5. Comprehensive privacy policy covering retention and user rights.

Q1 - Desired Behavior (R5)

The system shall store user preferences in chrome.storage.local for personalized experience. Shall provide settings page to view and manage stored data. Shall display welcome/consent screen on first install.

Q2 - Preventative Behavior (R5)

The system shall not transmit user search history or preferences to external servers. Shall not use chrome.storage.sync. Shall not collect location, browsing history, or cookies. Shall not retain data longer than necessary.

Q3 - Responsive Behavior (R5)

When user clicks 'Clear All Data', immediately delete all keys from chrome.storage.local, reset UI preferences, and show confirmation. On uninstall, automatically purge all storage. If consent withdrawn, disable data-dependent features.

Functional Requirements - Q1 (R5)

- FR-5.1.1: StorageModule shall use exclusively chrome.storage.local
- FR-5.1.2: PreferenceManagerModule shall store default_stopover_days, preferred_filter, and recent_searches
- FR-5.1.3: OptionsPageModule shall render options.html with settings controls
- FR-5.1.4: WelcomePageModule shall display onboarding.html with privacy explanation and consent checkbox

- FR-5.1.5: DataViewerModule shall display read-only table of stored data in options page.

Preventative Requirements - Q2 (R5)

- FR-5.2.1: NetworkPrivacyModule shall ensure ProxyServer receives only anonymous flight parameters with no user ID or IP logging
- FR-5.2.2: StorageScopeModule shall prohibit chrome.storage.sync, localStorage, sessionStorage, or cookies
- FR-5.2.3: DataMinimizationModule shall limit stored fields to those strictly necessary
- FR-5.2.4: CacheExpiryModule shall enforce automatic deletion of flight cache after 4 hours
- FR-5.2.5: ConsentGateModule shall disable data storage until user accepts privacy policy.

Responsive Requirements - Q3 (R5)

- FR-5.3.1: DataDeletionModule shall execute chrome.storage.local.clear() on 'Clear All Data' click
- FR-5.3.2: UIFeedbackModule shall display 'All data cleared' toast notification
- FR-5.3.3: UninstallCleanupModule shall register uninstall URL guiding users to feedback form
- FR-5.3.4: ConsentWithdrawalModule shall detect consent removal and disable personalization
- FR-5.3.5: AuditTrailModule shall log all deletion events with timestamp and reason
- FR-5.3.6: RecoveryInfoModule shall inform users deletion is irreversible.

6.6. Risk R6: Feature Overload and Scope Creep Exceeding Semester Capacity

Capability: Project Delivery & Scope Management

Risk Description: LayoverNow has an ambitious feature set for a solo developer in one semester. Attempting to implement all features risks burnout, missed deadlines, and a shallow, buggy MVP.

Likelihood	Impact	Risk Score	Priority
5	3	15	High

Justification for Likelihood: The project includes 6+ major features for a single developer in 8 weeks. Each feature could consume 2-3 weeks. Enthusiasm often leads to underestimating complexity. 80% of similar student projects experience scope issues.

Justification for Impact: Scope creep won't cause technical failure but results in a rushed, incomplete product. The demo may show half-implemented features. Grade impact is moderate if core functionality works, but learning and portfolio quality suffer.

Mitigation Strategy: 1. Strict MVP using MoSCoW: Must (flight search + deals + basic map), Should (AI chatbot with fallback), Could (visa/weather/currency), Won't (cross-browser, mobile app). 2. PRD with frozen scope by Week 2. 3. 1-week sprints with clear deliverables. 4. 1-week buffer before demo for bug fixes. 5. Track velocity; cut Could-features if sprints are missed.

Q1 - Desired Behavior (R6)

The system shall deliver an MVP within 8 weeks: flight stopover search with API, interactive radar map, deal filtering, and AeroAI chatbot with static fallback. Shall be installable from Chrome Web Store.

Q2 - Preventative Behavior (R6)

The system shall not add new features after Week 5 freeze without removing equivalent-effort feature. Shall not pursue cross-browser support, mobile responsiveness, or backend accounts. Shall not integrate hotel/restaurant APIs.

Q3 - Responsive Behavior (R6)

When sprint goal is missed or velocity drops below 50% for two consecutive sprints, automatically trigger scope reduction: defer all Could-priority features, simplify Should-priority to minimum form. Update PRD and GitHub Issues.

Functional Requirements - Q1 (R6)

- FR-6.1.1: MVPDefinitionModule shall document MVP scope in PRD.md with MoSCoW lists
- FR-6.1.2: SprintPlannerModule shall create GitHub Issues with story points for 1-week sprints

- FR-6.1.3: FlightSearchModule shall implement input, API query, and deal list as highest priority
- FR-6.1.4: RadarMapModule shall implement basic route plotting as Must feature
- FR-6.1.5: DealFilterModule shall implement three sorting algorithms as Must
- FR-6.1.6: AeroAIModule shall implement chat with static fallback as Should, after all Must features pass.

Preventative Requirements - Q2 (R6)

- FR-6.2.1: ScopeFreezeModule shall enforce code freeze on new features at end of Week 5
- FR-6.2.2: FeatureGateModule shall require scope review before any new feature is added
- FR-6.2.3: CrossBrowserBlockerModule shall reject issues targeting Firefox, Safari, or Edge
- FR-6.2.4: APIGateModule shall reject hotel/restaurant API integration without revised timeline
- FR-6.2.5: MobileBlockerModule shall defer mobile/responsive UI to post-semester
- FR-6.2.6: ComplexityAuditorModule shall flag features exceeding 2x original estimate for scope review.

Responsive Requirements - Q3 (R6)

- FR-6.3.1: VelocityTrackerModule shall calculate sprint velocity as completed/planned story points
- FR-6.3.2: ScopeReductionTriggerModule shall move all Could-priority Issues to 'Post-Semester' milestone when velocity <50% for two sprints
- FR-6.3.3: SimplificationEngineModule shall reduce Should-priority to minimum form
- FR-6.3.4: PRDUpdaterModule shall revise PRD.md within 24 hours of scope reduction
- FR-6.3.5: StakeholderNotificationModule shall update README and demo script to reflect current scope
- FR-6.3.6: BufferEnforcerModule shall reserve Week 7 for bug fixes and demo rehearsal only.

7. Week-by-Week Risk Tracking

The following table shows how each risk was re-analyzed, updated, and mitigated over the 8-week semester. New risks discovered during the project are included, and retired risks are documented with rationale.

Week	Phase	Active Risks	Total Exposure	Avg Score	Key Events
Week 1	Project Initiation	6 active	84	14.0	6 new risks identified at kickoff. R1 (API failure) highest at 20.
Week 2	Architecture Validation	8 active	90	11.25	R7 (CORS) and R8 (CWS rejection) discovered during prototyping. R1 impact reduced.
Week 3	Core Development	9 active	101	11.2	R9 (privacy) discovered during storage implementation. R2 probability reduced after offscreen doc success. R6 probability increased after budget laptop testing.
Week 4	Integration & Alpha	9 active	88	9.8	R8 RETIRED - extension approved in 36 hours. R10 (solo bandwidth) added. R1, R4, R5 scores reduced.

Week 5	Feature Completion	10 active	87	8.7	R11 (stale deals) discovered during user testing. Multiple risks trending down. Feature freeze declared.
Week 6	Beta Testing	11 active	72	6.5	R12 (cross-browser) added by beta tester request. R4 score drops to 4. Most risks now Medium or Low.
Week 7	Pre-Demo Hardening	11 active	61	5.5	R12 RETIRED - Chrome-only scope confirmed. R13 (post-launch adoption) added. R4 drops to 2.
Week 8	Final Demo	11 active (6 mitigated)	56	5.1	R14 (maintenance) added for post-semester. 6 risks fully mitigated. 5 residual risks accepted. Project delivered on time.

8. Retired Risks Log

R8 - Chrome Web Store Review Rejection

Category: External | Final Score: 2 (Low) | Retired: Week 4

Reason: Extension submitted with minimal permissions (activeTab, storage) and detailed privacy policy. Approved in 36 hours with zero rejections. Risk no longer relevant to project timeline.

Lesson: Early submission (Week 4) provided a 4-week buffer before final demo. Minimal permission model was key to fast approval.

R12 - Cross-Browser Compatibility

Category: Technical | Final Score: 2 (Low) | Retired: Week 7

Reason: Semester deliverable scope explicitly limited to Chrome. Edge compatibility noted as post-semester enhancement. Firefox/Safari builds require manifest divergence that exceeds semester capacity.

Lesson: Explicit scope boundaries in the PRD prevented wasted effort on non-essential platforms. Cross-browser support documented in README as future work.

9. Conclusion & Lessons Learned

The risk management process for LayoverNow demonstrated the value of early and continuous risk identification. Starting with 6 risks in Week 1 and growing to 11 by Week 8, the project maintained a downward trend in average risk score (from 14.0 to 5.1) despite discovering new risks throughout the semester.

Key lessons learned:

1. Early API research (Week 1-2) revealed CORS and rate-limiting issues before they became blockers. The proxy server architecture was the most important technical decision.
2. The MoSCoW prioritization and feature freeze (Week 5) prevented scope creep that would have derailed the project.
3. Submitting to the Chrome Web Store early (Week 4) eliminated a major schedule risk and provided confidence for the final demo.
4. Graceful degradation design (mock data for APIs, static tips for AI, Low Power Mode for canvas) turned potential failure points into resilient features.
5. Privacy-by-design from Week 1 prevented compliance issues and simplified the Web Store review process.
6. Solo developer bandwidth was the most persistent risk; buffer days and detailed documentation were essential for maintaining velocity through midterms.

The risk management process directly contributed to the successful on-time delivery of LayoverNow. All Must-have MVP features were implemented, the extension passed Chrome Web Store review, and the final demo was delivered without critical failures. The residual risks (R1, R5, R6, R11, R13) are accepted as inherent to the domain and are managed through monitoring and documented fallback procedures.